Antonio Mata Marín

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LinkedIn | Portfolio

Technical Artist specialized in improving real-time simulation workflows through tools, procedural systems, automation, and rendering technologies.

WORK EXPERIENCE

Technical Artist, CARLA Simulator

February 2022 – June 2026

- Developed **procedural content generation systems** and artist-facing tools for the **Digital Twins project**, including building generation, traffic sign generation, and **Unreal Engine editor extensions** that improved content creation workflows.
- Developed the **Primary Flight Display API** and **procedural visualization systems** for AeroSim, an aerospace simulation platform built with **Rust** and Unreal Engine in partnership with **Supernal**.
- Improved **rendering quality** and **production efficiency** by developing optimized cross-platform shaders and profiling rendering pipelines to identify and resolve performance bottlenecks.
- Extended **Isaac Sim** with **custom Python tooling** and **MDL materials**, improving workflow automation and expanding simulation capabilities.

EDUCATION

Implika, Obicex Barcelona.

- High Level degree on 2D and 3D videogames creation and Augmented/Virtual reality | 2018 - 2020

Implika, CampusNet Barcelona.

- High level degree on web programming and design| 2020 - 2022

Art College. Seville.

- One year of studies focused on art, sculpture, design, and perspective. 2023

REFERENCES

- **German Ros:** CARLA Project Lead; Director, Simulation Ecosystem Development. Linkedin
- **Marcos Delgado:** Art team lead (Ex-Manager). Linkedin
- **Aaron Samaniego:** Programming team lead. Linkedin

Reference contact details available upon request.

RESEARCH PROJECTS

- Raymarching renderer in Unity: **Implemented** a **raymarching rendere**r using Signed Distance Functions (**SDFs**), enabling the creation of complex procedural 3D content.
- NPR Rendering: Developed **multiple non-photorealistic shader**s for Unity, Unreal Engine and Blender
- Light-aligned Cordinates: Developed a **light-aligned UV** coordinate system in **Unity for procedural effects such as cross-hatching.**

TECHNICAL SKILLS

- **CORE:**

Languages:C++, C#, Python.

Shader Languages:HLSL, GLSL, MDL.

Software used: Unreal Engine 4/5, Unity, Blender, Substance package.

AI Integration: Effective use of AI tools to reduce development time, improve iteration speed, and increase overall productivity using Claude Code, Codex and Copilot, as well as generative 3D models.

Tooling: Git, Linux, Windows

- **ADDITIONAL:**

Languages: Rust, JavaScript.

Software used: NVidia Isaac Sim, 3Ds Max, ZBrush.

SOFT SKILLS

- **Cross-team collaboration:** Experienced collaborating with cross-functional teams and facilitating communication across different areas to achieve shared goals.
- **Adaptability:** Quick to adapt to new technologies, processes, and responsibilities.
- **Clear communication:** Clear and effective communicator, developed through daily collaboration with technical teams, managers, and cross-functional stakeholders.
- **Accountability:** Committed to quality and delivery, taking responsibility for the outcomes of my work.
- **Problem-solving skills:** Comfortable working autonomously, identifying problems and finding solutions beyond my primary area of expertise.

LANGUAGES

Spanish — Native
English — Full professional proficiency